

Formelregister — Formula Register

after Heim / Dröscher (1985)

Index of the numbered formulae of section V (*Formelregister*, pp. 100–148) of *Einführung in Burkhard Heim Elementarstrukturen der Materie mit Begriffs- und Formelregister*, Burkhard Heim / Walter Droscher, Andreas Resch Verlag, Innsbruck, 1985, ISBN 3-85382-038-7.

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(*)

$$\operatorname{rot} \vec{\mu} \sim \alpha \dot{\vec{G}} + \sigma \vec{v}, \quad \operatorname{div} \vec{G} \sim \sigma(r), \quad \alpha > 0$$

(*a)

$$\operatorname{rot} \vec{G} \sim \beta \dot{\vec{\mu}} + (\sigma - \sigma_{(0)}) \vec{f}, \quad \operatorname{div} \vec{\mu} \sim \sigma - \sigma_{(0)}, \quad \beta \neq 0$$

(*b)

$$\operatorname{divgrad} \vec{p} + \alpha \beta \ddot{\vec{p}} = \vec{0}, \quad \omega^2 \alpha \beta = 1, \quad 0 < \omega < \infty$$

(1)

$$R_{ik} - \frac{1}{2} g_{ik} R \sim T_{ik}$$

(1a)

$$g_{ik} = \sum_{\beta=1}^3 g_{ik}^{(\beta)} \neq g_{ki}^*$$

(1b)

$$\ddot{x}^i + \Gamma_{km}^i \dot{x}^k \dot{x}^m = 0$$

(2)

$$R_{ik} \sim w \eta_{ik}, \quad w = \sqrt{-g}, \quad g = |g_{ik}|_4$$

(2a)

$$\eta_{ik} = \frac{\Delta N_{ik}}{\Delta \Omega}, \quad \Delta \Omega = w \Delta x^1 \Delta x^2 \Delta x^3 \Delta x^4$$

(3)

$$C_{(p)} \varphi_{km}^{(p)} = \lambda_{(p)}(k, m) \varphi_{km}^{(p)}$$

(3a)

$$\varphi_{k(m)}^{(m)} = 0$$

(4)

$$(x_1, x_2, x_3) = (x_1^*, x_2^*, x_3^*) \equiv R_3, \quad x_4 = i\epsilon, \\ x_5 = i\varepsilon, \quad x_6 = i\eta$$

(4a)

$$1 \leq k \leq 6, \quad \vec{x}_k = \vec{e}_k x_k, \quad \vec{e}_i \vec{e}_k = \delta_{ik}$$

(5)

$$\sum_{k=1}^6 \cos^2 \alpha_k = 1$$

(5a)

$$j \neq 4, \quad 2\alpha_j = \pi, \quad \cos \alpha_4 = \pm 1$$

(5b)

$$\varphi(v)-\varphi(u)=\pi/2$$

(5c)

$$\hat{C}\left(R_6\right)=\left(\begin{array}{cccccc}\cos \psi & 0 & 0 & \sin \psi & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ -\sin \psi & 0 & 0 & \cos \psi & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0\end{array}\right), \quad u \beta=v, \quad i \beta=\operatorname{tg} \psi,$$

$$\psi=-\psi^*$$

(6)

$$g_{ik}\left(x_1\ldots x_6\right)=g_{ki}^*$$

(7)

$$\left(\left\{\begin{array}{c}i\\ km\end{array}\right\}\right)_6=\widehat{\{\}},\quad C=\sum_{p=1}^6C_p,\quad \vec{\lambda}=\sum_{p=1}^6\vec{e}_p\lambda_p$$

(8)

$$C\widehat{=} \vec{\lambda} \times \widehat{\{}}$$

(8a)

$$\operatorname{sp}\widehat{\widehat{C}}=\widehat{\vec{\lambda}}$$

(8b)

$$\vec{\lambda} \| sp \widehat{\{}} \rightarrow \operatorname{grad}_6 \ln w$$

(9)

$$g_{ik}\left(g_{ik}^{(1)}\ldots g_{ik}^{(3)}\right)=g_{ki}^*,\quad g_{ik}^{(\mu)}\left(x_1\ldots x_6\right)\neq g_{ki}^{(\mu)*}$$

(9a)

$$\widehat{\}}=\sum_{\mu=1}^3\widehat{\{}}_{(\mu)}^+, \quad \sum_{\mu=1}^3\widehat{\{}}_{(\mu)}^-=\hat{0}$$

(10)

$$3\omega=4c,\quad \omega^2\alpha|\beta|=1$$

(10a)

$$64\pi\gamma\alpha=3,\quad c^2|\beta|=12\pi\gamma$$

(10b)

$$8\pi\gamma\alpha'=1,\quad c^2\,|\beta'|=8\pi\gamma,\quad \omega'=c$$

(10c)

$$\alpha\operatorname{div}\vec{G}=\Lambda m(r),\quad 16V\Lambda=\left[3V\frac{d}{dV}+1\right]^2$$

(11)

$$rqe^{-q}=A(1-r/\varsigma)^2,\quad 16c^2A=3\gamma M_o,\quad M=Lm$$

$$1\leqslant L<\infty$$

(11a)

$$q=1-\sqrt{1-\varepsilon\varphi},\quad 8c^2\varepsilon=3$$

(11b)

$$r\varphi(r)=\gamma\operatorname{Lm}(r)(1-r/\varsigma)^2$$

(11c)

$$r^2G=-\gamma M_0\left(1-(r/\varsigma)^2\right)e^q(1+q)^{-1}$$

(12)

$$Y_1\varsigma\left(1-h\left(2Y_1m_0c\varsigma\right)^{-1}\right)^2=\frac{h^2}{\gamma m_0^3}. \quad 2Y_1m_0c\varsigma>h$$

(12a)

$$\gamma m_0^3 Y_1 \varsigma = h^2, \quad 2Y_1 m_o c \varsigma \not\geq h$$

(12b)

$$A_t^3\varsigma\approx 46[Mpc]$$

(13)

$$\begin{array}{lll} \varphi(r<\varrho)>O, & G(r<\varrho)<O, & \varphi(\varrho)=O, \\ G(\varrho)=O, & \varphi(r>\varrho)>O, & G(r>\varrho)>O \end{array}$$

(14)

$$R_{\pm}=\varsigma(1+\alpha)\left(1\pm\sqrt{1-(1+\alpha)^{-2}}\right), \quad \alpha e\gamma M_0=2\omega c\varsigma$$

(14a)

$$R_+=2\alpha\varsigma, \quad 2\alpha R_-=\varsigma, \quad R_+R_-=\varsigma^2$$

(14b)

$$4\omega cR_- = e\gamma M_0$$

(14c)

$$\begin{array}{l} R_- < r \leq \varsigma, \quad (G \leq 0), \quad \varsigma < r \leq R_+ < \infty \\ (G > 0) \end{array}$$

(15)

$$\tau\omega c^2=\pi\gamma\hbar$$

(15b)

$$m/p=M\geq 1, \quad (M)MOD(1)=O$$

(16)

$$\begin{array}{l} \bar{x}_k=\bar{X}_k;n, \quad \bar{X}_k=\bar{e}_k\alpha_k(), \quad (\bar{e}_i\bar{e}_k)_6=\hat{E}, \\ \sum\limits_{k=1}^6v_{kk}=1, \quad \alpha_k=\sqrt{\tau}, \quad k\leq 3, \quad \alpha_k=i\sqrt{\tau}, \\ k>3, \quad \Delta=\tau^3 \end{array}$$

(16a)

$$\left|{}^2\bar{\tau}\right|_6=\Delta^2$$

(16b)

$$\Delta_{\pm}\hat{=}\pm^2\bar{\tau}, \quad \Delta_{\pm}\hat{=}R_{6(\pm)}, \quad R_{6(+)}+R_{6(-)}\equiv R_{6(0)}$$

(16c)

$$R_{6(0)}\equiv R_{6(0)}\left(x^{\underline{k}}\right)_1^6, \quad x^{\underline{k}}=C^{\underline{k}};n,$$

(17)

$$\begin{array}{l} \Gamma;\psi=\lambda\psi, \qquad \varphi=\psi;n, \quad S_{\Omega}(\varphi^*\Gamma;\varphi-\varphi(\Gamma;\varphi)^*)\delta\Omega=0, \\ S_{\Omega}\varphi\varphi^*\partial\Omega<\infty, \quad \lambda=\lambda^* \end{array}$$

(18a)

$$\begin{array}{l} R_{kmp}^i\rightarrow \varsigma_{kmp}^i;n, \quad {}^2\bar{\varsigma}=sp^4\bar{\varsigma}, \quad \varsigma=sp^2\bar{\varsigma} \\ sp()_{(-)}^{(-,-)};\left({}^2\bar{\varsigma}-\frac{1}{2}{}^2\bar{\gamma};()\varsigma\right)=\bar{0} \end{array}$$

(19)

$$L;\widehat{\square}={}^4\overline{0}$$

(19a)

$$L = K - \bar{\lambda} \times ()$$

(19b)

$$\mathrm{sp}\, K; \widehat{\square} = \widehat{\lambda \square}$$

(19c)

$$\bar{\lambda} \perp ()_{(-)}; \varphi, \quad \varphi; n = \ln \sqrt{-g}$$

(20)

$$(E-\psi_{kl})^{\Lambda_{kl}+1}\cdot\psi_{kl}^{\Lambda_{kl}-1}=2^{-2\Lambda_{kl}}\cdot C_{kl}e^{-\lambda_{kl}\mu},$$

$$\psi_{kl}=\frac{b_i^{(kl)}}{\lambda(kl)}\left[\begin{array}{c}i\\kl\end{array}\right],\quad \mu=\sum_{m=1}^q\alpha_m()_m,\quad C_{kl}=\mathrm{const}$$

(20a)

$$\Lambda_{kl}=\alpha_l(a(kl)-1)^{-1}-\sum_{m\neq l}\alpha_m$$

$$(a(kl)-q)\cdot\lambda_{kl}=\lambda(kl),$$

$$\left(\lambda(kl),a(kl),b_i^{(kl)}\right)=\sum_{m=1}^q\left(\lambda_m(k,l),a_m(kl),b_{mi}^{(kl)}\right),$$

$$b_{mi}^{(kl)}=\left(\frac{a_{li}}{a_{lk}}-\frac{a_{mi}}{a_{mk}}\right)\frac{a_{km}}{a_{kl}},\quad a_m^{(kl)}=\frac{a_{km}}{a_{kl}},\quad \lambda_m(\tilde{k},l)=\lambda_m(k,\tilde{l})=$$

$$a_{ml}\lambda_m(m,l)=-\lambda_l(m,m),\quad \lambda_m(k,m)=\lambda_m(m,m)=0$$

$$=\lambda_m(\tilde{k},\tilde{l})=\lambda_{\tilde{m}}(k,l)=0,$$

(20b)

$$\varepsilon_+^{(\pm)}=\pm\frac{\pi}{2}\left(2(\cdot)_++E\right),\quad \varepsilon_-^{(\pm)}=\pm\pi(\cdot)_-,$$

$$n_{\pm}\geq 0,\quad \beta_+^{(\pm)}+\beta_-^{(\mp)}=\pm\frac{\pi}{2},$$

(21)

$$\sum_{k=1}^q\gamma_{ik}=A_i\exp\left(\sum_{k,l=1}^qc_{kl}^{(i)}S\psi_{kl}\partial\mu\right),\quad c_{kl}^{(i)}b_i^{(kl)}=2\lambda(kl).$$

(21a)

$$\left[\begin{array}{cc}\tilde{r}&\\k&l\end{array}\right]=B_{\tilde{k}l}^{\tilde{r}}\cdot\exp\lambda_{kl}\left(\mu-\mathbf{S}\psi_{kl}\partial\mu\right)$$

(21b)

$$\gamma_{kl}^{(w)}\equiv\psi_{kl}^{(\min)}=0,\quad \psi_{kl}^{(\max)}=E$$

(21c)

$$\underline{\gamma}_{ii}=A_i\exp\left(\sum_{k,l=1}^qc_{kl}^{(i)}S\underline{\psi}_{kl};() \partial \underline{\mu}\right),\quad \hat{S}^2\bar{\gamma}_q\hat{S}^\times=^2\bar{\gamma}_-$$

$$\hat{S}\hat{S}^\times=\hat{E},\quad \hat{\beta}\hat{\beta}^\times=\hat{E},\quad \underline{\mu}=\sum_{k=1}^q\beta_k()_k,$$

$$\beta_k=\sum_{i=1}^q\alpha_i\beta_{ik}$$

(22)

$$\psi_{kl}\approx\left(E+C_{kl}\cdot e^{-\lambda_{kl}\mu}\right)^{-1},\quad \Lambda_{kl}\approx 0$$

(22a)

$$\tau\rightarrow 0\psi_{kl}\left(x_s\right)_1^q=\left(1+C_{kl}e^{\pm i\lambda_{kl}y}\right)^{-1},\quad y^2=\varsigma^2-r^2,$$

$$-\varsigma^2=\sum_{s=4}^qx_s^2,\quad r^2=\sum_{i=1}^3x_i^2$$

(22b)

$$r^2 \gg \varsigma^2, \quad \psi = (1 + Ce^{\pm \lambda r})^{-1}$$

(23)

$$\begin{aligned} H_{(1)} &= \pi \underline{\varepsilon} n_1, \quad H_{(2)} = \pi \underline{\varepsilon} n_2, \\ -\varsigma^2 &= x_5^2 + x_6^2 = -\varepsilon^2 - \eta^2, \quad \leq \underline{\varepsilon} = \text{const} > 0 \end{aligned}$$

(24)

$$a \equiv G\left(x_5,x_6\right) \rightleftharpoons P\left(x_4,x_5,x_6\right) \equiv b$$

(25)

$$r=\frac{2n_r+1}{2n_\varsigma+1}\cdot\varsigma$$

(26)

$$m=2\sqrt{\frac{ch}{\gamma}\frac{\sqrt[4]{2n}}{\sqrt{2n-1}}}$$

(26a)

$$\begin{aligned} \eta &= \lambda/2, \quad r = \varepsilon = \frac{\Lambda}{2}, \quad \lambda = \frac{h}{mc}, \\ \Lambda &= \frac{h}{\mu\omega} = 2\varsigma, \quad \varsigma = \frac{h^2}{\gamma m^3} \end{aligned}$$

(26b)

$$n=n(t), \quad \frac{dn}{dt} \geq 0$$

(27)

$$m(n,q)=m(n)\eta_q, \quad \eta_q\sqrt[4]{4q^4+\pi^4}=\pi$$

(27a)

$$m(n)=\mu\frac{\sqrt[4]{2n}}{\sqrt{2n-1}}, \quad \mu=\sqrt{ch/\gamma}$$

(27b)

$$a\pi^2\varepsilon_{\pm}=\pm3\sqrt{2\hbar/R_-}, \quad R_-=\sqrt{\mu_0/\varepsilon_0}, \quad (a=\sqrt{2})$$

(27c)

$$a\pi^2C_{\pm}=\pm\sqrt{2\hbar/R_-}, \quad \varepsilon_{\pm}=3C_{\pm}$$

(28)

$$V_{xy}=e_xe_yf(r)$$

(28a)

$$\begin{aligned} e_r &= \varepsilon_{\pm} \sqrt{\eta}, \quad e_d = \varepsilon_{\pm} (1 - \sqrt{\eta}), \quad 2e_w = \varepsilon_{\pm} (1 + \sqrt{\eta}), \\ \eta \sqrt[4]{4 + \pi^4} &= \pi \end{aligned}$$

(29)

$$4\pi^2e_{\pm}=\pm3\sqrt{2\vartheta\hbar/R_-}, \quad \vartheta=5\eta+2\sqrt{\eta}+1$$

(29a)

$$(2\pi)^5\alpha'=9\vartheta$$

(30)

$$eR\sqrt{\tau}=\varepsilon nF, \quad \varepsilon^2\sqrt{F}=s_0=1[m]$$

(31)

$$\begin{aligned} m_q &\approx \frac{\hbar}{\beta_c} - \sqrt[3]{4\alpha_q/\varepsilon}, \quad \beta = \sqrt{\tau\sqrt{\pi}}, \quad \varepsilon\tau = E, \\ \alpha_q &= 1 + 4\left(\frac{q}{\pi}\right)^4 \end{aligned}$$

(32)

$$m_L c s_0 = 4 \sqrt[4]{\pi} \sqrt[3]{3 \pi \gamma \hbar s_0} \sqrt{\frac{c \hbar}{3}}, \quad m_e \eta \sqrt[3]{\eta} = m_L, \\ m_B \eta^2 \sqrt[3]{\eta} = m_L, \quad m_L < m_e < m_B, \quad s_0 = 1[m]$$

(33)

$$m_c (1 - \eta_q) > q m_e$$

(34)

$$\underline{\eta} = \sqrt[3]{\tau \varepsilon / 3}$$

(34a)

$$p \underline{\eta} = \pi \hbar, \quad p = \frac{\hbar}{2} \left(\frac{\tau}{3} \cdot \underline{\varepsilon} \right)^{-1/3}, \\ \frac{w}{2p} \geq \underline{\eta}, \quad w = N \hbar$$

(35)

$$s' = b \tau^{5/6}, \quad b = \frac{e E^{-1/3}}{\sqrt[4]{\pi} \sqrt[3]{2}}$$

(35a)

$$\vartheta = \frac{3b}{4c} \tau^{5/6}, \quad 2\omega c^2 \tau = \gamma \hbar, \quad \gamma \mu^2 = c \hbar, \\ \pi^4 \kappa^2 R_- = 9 \hbar$$

(36)

$$h = b \sqrt{3/2} \frac{\mu}{\vartheta} \tau^{4/3}, \quad \gamma = \frac{3}{4} \sqrt{3/2} (b \tau / \vartheta)^2 \cdot \frac{\sqrt[6]{\tau}}{\mu} \\ \varepsilon_o = (\pi \kappa)^2 \left(\frac{2}{3} \pi \right)^3 \sqrt{2/3} \left(\frac{\vartheta}{b \tau} \right)^2 (\mu \sqrt[6]{\tau})^{-1} \\ \mu_o = \frac{6}{\pi^5} \sqrt{3/2} \frac{\mu \sqrt{\tau}}{\kappa^2}$$

(37)

$$f \sqrt{3/2} \left(\frac{D}{4 \sqrt{\tau}} f^3 \sqrt{3/2} - 1 \right)^2 = \frac{D}{\sqrt{\tau}}$$

(37a)

$$\left(\frac{e D \sqrt{\tau}}{\pi E} - 1 \right) f^2 = \sqrt{\frac{e D \sqrt{\tau}}{\pi E}}, \quad E = 1 [m^2]$$

(37b)

$$a D \approx \tau^{-11/6}, \quad 2a = (3\pi 2^5)^{-2} E^{-7/3}$$

(38)

$$DA = \dot{D}, \quad A \neq 0, \quad A = 0$$

(39)

$$h \sim D^{-8/11}, \quad \gamma \sim D^{-13/11}, \quad \varepsilon_0 \sim D^{13/11}, \\ \mu_0 \sim D^{-3/11}$$

(39a)

$$D\dot{y} = n\dot{D}y$$

(40)

$$3W = 16\pi \left(\frac{a}{b} \right)^2 \sqrt[3]{3\tau^2}, \quad a^2 N^3 \sqrt{3\tau^2} = 2\pi \\ 3\alpha' NW = (4\pi)^2$$

(40a)

$$a s_0 \eta \sqrt[3]{2\eta} = 4 \sqrt[4]{\pi} \sqrt[3]{3 s_0}, \quad s_0 = 1[\text{ m}], \\ 4\pi^2 b = 3 \sqrt{\vartheta/\pi}, \quad \vartheta = 5\eta + 2\sqrt{\eta} + 1$$

(41)

$$DA = \dot{D} > 0, \quad \dot{\tau} < 0$$

(41a)

$$A \ll H, \quad z \approx Bs, \quad v \approx Hs, \quad H = cB$$

(42)

$$cz_D = As, \quad z_S = -\frac{39\lambda}{11c}A, \quad z_a = \frac{16\tau}{9\lambda^2}$$

(42a)

$$z_k = z_D + z_S + z_a$$

(43)

$$D(T) = D_0 e^\psi, \quad \dot{\psi} = A, \quad D_0 > 0$$

(43a)

$$\frac{d}{dt} \ln m = \frac{36}{11} A > 0$$

(44)

$$\lambda_r^3 \approx \frac{c\tau}{2A}$$

(44a)

$$4\lambda_r^3 \approx \tau D, \quad c\lambda_r^3 = \text{const}, \quad \frac{\sqrt{h}}{m_r^3} = \text{const},$$

$$\left(\frac{\lambda_r}{m_r}\right)^3 c\sqrt{h} = \text{const}$$

(45)

$$z \approx Bs(1 - Bs)^{-1} > 0, \quad cB = \sqrt{\pi e \gamma \sigma}, \quad s \geq \rho,$$

$$\gamma m^3 \varrho = h^2, \quad z\lambda \sim \delta\lambda$$

(45a)

$$H \approx \sqrt{\pi e \gamma \sigma}$$

(46)

$$y\sqrt{\pi e \gamma \sigma} \approx c$$

(46a)

$$s < y, \quad 2y \ll D$$

(46b)

$$s \leq \varrho, \quad z = 0, \quad \varrho < s < y, \quad z > 0$$

$$s \rightarrow y, \quad z \rightarrow +\infty$$

(47)

$$\eta^7 - \eta = \pm a, \quad 2\eta^2 = f_{(0)} \sqrt[6]{6/\pi}, \quad a\sqrt{\pi} = \sqrt[6]{\pi/6}$$

$$f_{(0)} \left(eD_0^2 (E\sqrt{\pi})^{-1} - 1 \right)^{1/2} = \sqrt[4]{eD_0^2 (E\sqrt{\pi})^{-1}}$$

(48)

$$D_f(t=0) < D_m(t=0) < D_p(t=0)$$

(48a)

$$D_f = 0,90991798, D_m = 1,06425809, D_p = 3,70121160$$

(49)

$$T_1^+ \left(D_{fmp}^{xy+} \right) \frac{D(R_3^+)}{\cos \alpha_4 > 0} \rightarrow T_2^+ \left(D_{fmp}^{yx-} \right) \equiv$$

$$\equiv T_1^- \xrightarrow[\cos \alpha_4 < 0]{D(R_3^-)} T_2^- \equiv T_1^+$$

(49a)

$$\begin{aligned} D_0 &\leq D, \quad A > 0, \quad \dot{\tau} < 0, \quad D_{\max} = \Lambda < \infty, \\ A &= 0, \quad \dot{\tau} = 0, \quad A > D \geq D_e > 0, \quad A < 0, \\ \dot{\tau} &> 0 \end{aligned}$$

(50)

$$\vec{x}_5^{(g)} + \vec{x}_6^{(g)} = i\vec{M} = \text{const}(x_4) < \infty, \quad \dot{\varepsilon}_g^2 + \dot{\eta}_g^2 = 0$$

(50a)

$$T_0(D_{fmp}) = 0 \leq t \leq \theta(D_{pmf}) = 2T_A < \infty$$

(51)

$$D\ddot{D} - a\dot{D}^2 = 0, \quad 11a = -5$$

(51a)

$$D^\beta - D_1^\beta = 2\beta C(t - t_1), \quad \beta = 1 - a$$

(52)

$$\left(\frac{A}{D}\right)^{3/11} \sqrt{E} = 2^9 e^4 \sqrt{\pi} \left(9\pi^2 \sqrt[3]{2}\right)^{-1} \left(\frac{\tau}{E}\right)^{11/6} D$$

(53)

$$\begin{aligned} 1 \leq (\mu, v) \leq 3, \quad {}^2\bar{\kappa}_{(\mu)} \neq {}^2\bar{\kappa}_{(\mu)}^\times, \quad {}^2\bar{\gamma}_{(\mu\nu)} = sp({}^2\bar{\kappa}_{(\mu)} \times {}^2\bar{\kappa}_{(v)}) \\ \hat{\gamma} = ({}^2\bar{\gamma}_{(\mu\nu)})_3 \end{aligned}$$

(53a)

$$\begin{aligned} {}^2\bar{\kappa}_{(1)} &= {}^2\bar{\kappa}_{(1)}(C_5, C_6), \quad {}^2\bar{\kappa}_{(2)} = {}^2\bar{\kappa}_{(2)}(C_4), \\ {}^2\bar{\kappa}_{(3)} &= {}^2\bar{\kappa}_{(3)}(C_1, C_2, C_3) \end{aligned}$$

(54)

$$\begin{aligned} S(k); S(i); {}^2\bar{\gamma} = {}^2\bar{\gamma}({}^2\bar{\gamma}_{(\nu\nu)}, {}^2\bar{\kappa}_{(\nu)}), \quad i \neq k \neq \nu, \\ \hat{\gamma} \neq \hat{\gamma}^\times, \quad S(k); S(i); {}^2\bar{\gamma} = {}^2\bar{\gamma}_{(\mu\nu)+}, \quad \hat{\gamma} = ({}^2\bar{\gamma}_{(\mu\nu)} \cdot \delta_{\mu\nu})_3, \\ \bar{\psi}_{(\mu)} = S^2\bar{\kappa}_{(\mu)}; ()\partial\bar{x} \neq \bar{\psi}_{(\mu)}^\times \end{aligned}$$

(54a)

$$\begin{aligned} {}^2\bar{\kappa}({}^2\bar{\kappa}_{(\mu)})_1^3 &= {}^2\bar{\kappa}^\times, \quad \bar{\psi}(\bar{\psi}_{(\mu)})_1^3 = \bar{\psi}^\times, \\ \widehat{\mathbb{I}} &= f\left(\left[\begin{array}{c} \widehat{\kappa\lambda} \\ -\mu\nu \end{array}\right]\right)_1^3 = \widehat{\mathbb{I}}^\times, \quad {}^2\bar{\kappa}_{(\mu)} \neq {}^2\bar{\kappa}_{(\mu)}^\times, \\ \left[\begin{array}{c} \widehat{\kappa\lambda} \\ -\mu\nu \end{array}\right] &\neq \left[\begin{array}{c} \widehat{\kappa\lambda} \\ -\mu\nu \end{array}\right] \end{aligned}$$

(55)

$$\begin{aligned} {}^2\bar{\gamma}_{(pq)+} &= \text{const}(Z_{\bar{k}}), \quad \partial_{\bar{k}}\gamma_{-il}^{(pq)} = \frac{\alpha_{\bar{k}}}{\alpha_i}\partial_i\gamma_{\bar{k}l}^{(pq)} - \\ &- \frac{\alpha_{\bar{k}}}{\alpha_l}\partial_l\gamma_{\bar{k}i}^{(pq)}, \quad {}^2\bar{\gamma}_{(pq)-} \equiv f(Z_{\bar{k}}) \end{aligned}$$

(55a)

$${}^2\bar{\gamma}_{(pq)-} = {}^2\bar{P}_{(pq)}, \quad \tau\alpha_m\gamma_{+\tilde{k}m}^{(pq)} = \partial_m\phi_{\tilde{k}}^{(pq)}$$

(56)

$$\begin{aligned} \gamma_{(\mu\nu)+}^{\bar{k}l} &= \text{const}, \quad \gamma_{(\mu\nu)+}^{kl} = \gamma_{(\mu\nu)+}^{kl}(V_r), \quad \gamma_{-\bar{k}l}^{(\mu\nu)} = 0, \\ \gamma_{-kl}^{(\mu\nu)} &= \text{const} \neq 0 \end{aligned}$$

(56a)

$$\left[\begin{array}{c} \widehat{\kappa\lambda} \\ -t \\ \mu\nu \end{array}\right] = \left[\begin{array}{c} \widehat{\kappa\lambda} \\ -t \\ \mu\nu \end{array}\right]_+ = \left[\begin{array}{c} \widehat{\kappa\lambda} \\ -t \\ \mu\nu \end{array}\right] x$$

(57)

$$\widehat{\mathbb{I}} = \sum_{\varepsilon} \left(\left[\begin{array}{c} \widehat{\kappa\lambda} \\ -+ \\ \mu\nu \end{array} \right]_{+} + sp^2 Q_{(\mu\nu)}^{(\kappa\lambda)}; () \times \left[\begin{array}{c} \widehat{\kappa\lambda} \\ -+ \\ \mu\nu \end{array} \right]_{+} \right),$$

$$\varepsilon \equiv (\kappa, \lambda, \mu, \nu)$$

(58)

$$L_{(\mu\nu)}^{(\kappa\lambda)}; \left[\begin{array}{c} \widehat{\kappa\lambda} \\ -+ \\ \mu\nu \end{array} \right] = {}^4\overline{0}, \quad \left[\begin{array}{c} \widehat{\kappa\lambda} \\ -+ \\ \mu\nu \end{array} \right] = \left[\begin{array}{c} \widehat{\kappa\lambda} \\ -^{\circ}+ \\ \mu\nu \end{array} \right]_{+}$$

(58a)

$$\begin{aligned} & + \delta(\underline{k}, s); () \sum_{\varepsilon} Q_{(\alpha\beta)t}^{(\gamma\delta)\underline{s}} \left[\begin{array}{c} t \\ km \end{array} \right]_{(\alpha\beta)}^{(\gamma\delta)} - \\ & - \delta(\underline{kl}, sm); () \sum_{\varepsilon} Q_{(\alpha\beta)t}^{(\gamma\delta)\underline{s}} \left[\begin{array}{c} t \\ k'm \end{array} \right]_{(\alpha\beta)}^{(\gamma\delta)} + \\ & + Q_{(\mu\nu)t}^{(\kappa\lambda)\underline{i}} \delta(\underline{ik}, ts); () \sum_{\varepsilon} \left[\begin{array}{c} s \\ km \end{array} \right]_{(\alpha\beta)}^{(\gamma\delta)} (1 - \delta) - \\ & - Q_{(\mu\nu)t}^{(\kappa\lambda)\underline{i}} \delta(\underline{ikl}, tsm); () \sum_{\varepsilon} \left[\begin{array}{c} s \\ kl \end{array} \right]_{(\alpha\beta)}^{(\gamma\delta)} (1 - \delta) \\ & \delta = \delta_{\mu\alpha} \delta_{\nu\beta} \delta_{\kappa\gamma} \delta_{\lambda\delta}, \quad \delta(\underline{p} \dots \underline{q}, m \dots s); A_{p\dots q} = A_{m\dots s} \end{aligned}$$

(59)

$$\widehat{\lambda\mathbb{I}} = \sum_{\varepsilon} \bar{\lambda}_{(\mu\nu)}^{(\kappa\lambda)} \left[\begin{array}{c} \widehat{\kappa\lambda} \\ -^{\circ}+ \\ \mu\nu \end{array} \right], \quad \left[\begin{array}{c} \widehat{\kappa\lambda} \\ -^{\circ}+ \\ \mu\nu \end{array} \right] = \left(1 + sp^2 \bar{Q}_{(\mu\nu)}^{(\kappa\lambda)} \times () \right); \left[\begin{array}{c} \widehat{\kappa\lambda} \\ -\mu\nu \end{array} \right]$$

(60)

$$\left[\begin{array}{c} \widehat{\kappa\lambda} \\ -+ \\ \mu\nu \end{array} \right]_{-} = \hat{0}, \quad \left[\begin{array}{c} \kappa\lambda \\ -\mu\nu \end{array} \right]; () \left[\begin{array}{c} \widehat{\kappa\lambda} \\ -+ \\ \mu\nu \end{array} \right] \times; () \partial\Omega < \infty, \quad \overset{\mathbf{S}}{0} sp \left({}^2Q_{(\mu\nu)}^{(\kappa\lambda)} \times \left[\begin{array}{c} \widehat{\kappa\lambda} \\ -+ \\ \mu\nu \end{array} \right] \right); () \left(sp \left({}^2\bar{Q}_{(\mu\nu)}^{(\kappa\lambda)} \times \left[\begin{array}{c} \widehat{\kappa\lambda} \\ -+ \\ \mu\nu \end{array} \right] \right) \right) \overset{\mathbf{S}}{0}$$

$$\Lambda_{(\mu\nu)}^{(\kappa\lambda)} = \left(\Lambda_{(\mu\nu)}^{(\kappa\lambda)} \right) \times$$

(60a)

$$sp L_{(\mu\nu)}^{(\kappa\lambda)}; \left[\begin{array}{c} \widehat{\kappa\lambda} \\ -+ \\ \mu\nu \end{array} \right] = {}^2\overline{0}, \quad L_{(\mu\nu)}^{(\kappa\lambda)}; sp \left[\begin{array}{c} \widehat{\kappa\lambda} \\ -+ \\ \mu\nu \end{array} \right] = {}^2\overline{0}$$

(61)

$$L_{(\mu\nu)+}^{(\kappa\lambda)}; \left[\begin{array}{c} \widehat{\kappa\lambda} \\ -\mu\nu \end{array} \right]_{+} = {}^4\overline{0}$$

(61a)

$$\begin{aligned} L_{(\mu\nu)+}^{(\kappa\lambda)} &= \left(K_{(\mu\nu)+}^{(\kappa\lambda)} - \bar{\lambda}_{(\mu\nu)}^{(\kappa\lambda)} \times () \right); \left(1 + sp^2 \bar{Q}_{(\mu\nu)}^{(\kappa\lambda)} \times () \right) + D_{(\mu\nu)}^{(\kappa\lambda)}, \\ \Lambda_{(\mu\nu)+}^{(\kappa\lambda)} &= K_{(\mu\nu)+}^{(\kappa\lambda)}; \left(1 + sp^2 \bar{Q}_{(\mu\nu)}^{(\kappa\lambda)} \times () \right) + D_{(\mu\nu)}^{(\kappa\lambda)}, \\ K_{(\mu\nu)+m}^{(\kappa\lambda)} &= \left(\kappa\lambda + \right)_{(-)l}^{(+,-)}; \delta(l, m) - \left(\mu^+ \right)_{(-)m}^{+} \right)_{(-,-)}^{(+\lambda)} \end{aligned}$$

(62)

$$\begin{aligned} & \begin{bmatrix} i \\ kl \end{bmatrix} (\mu\lambda) + Q_{(\mu\nu)+}^{(\kappa\lambda)i} \begin{bmatrix} m \\ k \end{bmatrix} \begin{pmatrix} \kappa\lambda \\ (\mu\nu)+ \end{pmatrix} = \\ & = A_{(\mu\nu)kl}^{(\kappa\lambda)\frac{i}{kl}} \exp \left(\underline{\lambda}_{kl}\mu - \underline{c}_s \mathbf{S} \begin{bmatrix} s \\ kl \end{bmatrix} ; () \partial_\mu \right) \end{aligned}$$

(63)

$$\begin{aligned} & \begin{bmatrix} i \\ kl \end{bmatrix}_{(\mu\nu)+}^{(\kappa\lambda)} + Q_{(\mu\nu)m}^{(\kappa\lambda)} \begin{bmatrix} m \\ k \\ \underline{\alpha}_{kl} \end{bmatrix}_{(\mu\nu)+}^{(\kappa\lambda)} = C_{(\mu\nu)kl}^{(\kappa\lambda)i} e^{\lambda_{kl}\mu} (e^{\lambda_{kl}\mu} - E)^{-\underline{\alpha}_{kl}}, \\ & T_{(\mu\nu)}^{(\kappa\lambda)} = T \end{aligned}$$

(64)

$$\underline{\psi}_{kl} = \psi_{kl}^{\alpha_{kl}}, \quad \underline{\lambda}_{kl} = \underline{\alpha}_{kl} \lambda_{kl}$$

(65)

$$\beta_{-+}^{(\pm)} = \pm \frac{\pi}{2} (2()_{+} + E), \quad \beta_{--}^{(\pm)} = \pm \pi()$$

(65a)

$$\alpha_{kl} = n_{kl} (2n_{kl} + 1)^{-1}$$

(66)

$$\begin{aligned} & \partial \gamma_{+kl}^{(\mu\nu)} = 0, \quad \psi_{(\mu\nu)kl}^{(\kappa\lambda)} = 0, \quad \partial^2 \gamma_{+kl}^{(\mu\nu)} = 0, \\ & \partial q_{(\mu\nu)kl}^{(\kappa\lambda)} = 0, \quad \partial \psi_{(\mu\nu)kl}^{(\kappa\lambda)} = 0, \\ & q_{(\mu\nu)kl}^{(\kappa\lambda)} = C_{(\mu\nu)kl}^{(\kappa\lambda)} \sum_{s=1}^q \left(\sum_{i=1}^q \left(\gamma_{(\kappa\lambda)}^i \frac{S}{\lambda} + Q_{(\mu\nu)m}^{(\kappa\lambda)} \gamma_{(\kappa\lambda)}^m \frac{S}{\lambda} \right) \right) - 1 \end{aligned}$$

(66a)

$$q_{(\mu\nu)kl}^{(\kappa\lambda)} = q_{(\mu\nu)kl}^{(\alpha\beta)} \psi_{(\mu\nu)kl}^{(\alpha\beta)} \left(\psi_{(\mu\nu)kl}^{(\kappa\lambda)} \right)^{-1}$$

(67)

$$\begin{aligned} & sp_-^2 \bar{\gamma}_{(\mu\nu)} = C_{(\mu\nu)} \left| (e^{\lambda_{kl}\mu} - E)^{\underline{\alpha}_{kl}} \delta_{kl} \right|_q, \quad \lambda_{kl} = \lambda_{kl}^{(\mu\nu)}, \\ & \underline{a}_{kl} = a_{kl}^{(\mu\nu)} = \text{const} \end{aligned}$$

(68)

$$\begin{aligned} & Q_{(\mu\nu)}^{(\kappa\lambda)} = A_{(\mu\nu)}^{(\kappa\lambda)} \sum_{l=1}^q C_{(\mu\nu)l}^{(\kappa\lambda)} (E - e^{-\lambda_l \mu})^{-\underline{\alpha}_l} \cdot \left(\sum_{k=1}^q C_l^{(\mu\nu)} (E - \right. \\ & \left. - e^{-\lambda_l^{(\mu\nu)} \mu})^{-1} \right)^{-1} \cdot \left| \delta_{kl} (e^{\lambda_l^{(\kappa\lambda)} \mu} - E)^{\alpha_{kl}^{(\kappa\lambda)}} \right|_q \cdot \left| \delta_{kl} (e^{\lambda_l^{(\mu\nu)} \mu} - \right. \\ & \left. - E)^{\alpha_{kl}^{(\mu\nu)}} \right|_q^{-1} - qE \end{aligned}$$

(68a)

$$\begin{aligned} & \underline{\partial} Q_{(\mu\nu)}^{(\kappa\lambda)} = 0, \quad \underline{\partial} \psi_{ll} = 0, \quad \underline{\partial}^2 ({}^2 \bar{\gamma}_{(\kappa\lambda)}) = {}^2 \bar{0}, \\ & \underline{\partial}^2 ({}^2 \bar{\gamma}_{(\mu\nu)}) = {}^2 \bar{0} \end{aligned}$$

(68b)

$$Q_{(\mu\nu)\text{ext}}^{(\kappa\lambda)} \equiv \begin{pmatrix} \kappa\lambda \\ \mu\nu \end{pmatrix}$$

(69)

$$\begin{aligned} & \kappa'_{(1)} Q_- = Q_+ = C (e^{\lambda\mu} - E)^a (e^{\Lambda\mu} - E)^b, \\ & \hat{S}^2 \bar{\kappa}_{(1)} \hat{S}^\times = {}^2 \bar{\kappa}'_{(1)}, \quad \kappa'_{(1)ik} \sim \delta_{ik} \end{aligned}$$

(69a)

$$\begin{aligned}
Q_+(y) &= \lim_{\tau \rightarrow 0} Q_+; n = C (e^{\pm i\lambda y} - 1)^a, \\
\beta_{(a)}^+ &= \pm 2\pi n_{(a)}, \quad y^2 = \varepsilon^2 + \eta^2, \quad \beta_{(a)}^-(Q_-) \equiv \beta_{(a)}^+, \\
\kappa'_{(1)} &= \kappa_{\text{ext}}^{(1)}
\end{aligned}$$

(70)

$$\begin{aligned}
K_1^- &= 1, 4, 5. & K_2^- &= 2, 4, 6. & K_3^- &= 5, 3, 6. \\
K_1^- &= 1, 4. & K_2^- &= 1, 5. & K_3^- &= 2, 4. \\
K_4^- &= 2, 6. & K_5^- &= 3, 5. & K_6^- &= 3, 6, \\
(1 \dots 6)_- &= (6 \dots 1)_+
\end{aligned}$$

(70a)

$$\begin{aligned}
Q_1(\text{I}h \rightarrow \text{I}a) &= Q_1 (K_i^{++} \rightarrow K_i^{--})_1^3, \\
Q_1^{(o)} &= Q_1(1, 2, 3 \rightarrow 6, 5, 4), \\
Q_2^+(\text{I}h \rightarrow \text{II}a) &= Q_2 (1, 2, 3 \rightarrow K_{3,5}^- K_{1,6}^- K_{2,4}^-), \\
Q_2^-(\text{II}h \rightarrow \text{I}a) &= Q_2((1, 6)(2, 5)(3, 4) \rightarrow (4, 5)(4, 6)(5, 6)), \\
Q_3(\text{III}h \rightarrow \text{II}a) &= Q_3 (K_k^+ \rightarrow K_k^-)_1^6, \\
Q_4^+(\text{I}h \rightarrow \text{III}h) &= Q_4 (1, 2, 3 \rightarrow K_{5,3}^+, K_{6,1}^+ K_{4,2}^+), \\
Q_4^-(\text{II}a \rightarrow \text{I}a) &= Q_4 (K_{2,4}^-, K_{1,6}^-, K_{3,5}^- \rightarrow 4, 5, 6)
\end{aligned}$$

(71)

$$\begin{aligned}
K_{11}^- \left(\begin{array}{c} 2 \\ \gamma, \gamma^* \end{array} \right) &= 6d, 7s, 9s, 2e, 3e \\
C_{11}^- &= 1d, 3d, 2s, 5s \\
K_{12}^- \left(\begin{array}{c} 3 \\ \gamma, \gamma^* \end{array} \right) &= 6d, 7s, 8s, 2e, 3e \\
C_{12}^- &= 2d, 3d, 3s, 6s \\
K_{13}^- \left(\begin{array}{c} \alpha, \alpha^* \\ \beta, \beta^* \end{array} \right) &= 5d, 4s, 5s, 6s, 3e \\
C_{13}^- &= 4d, 1s, 2s, 3s, 2e \\
K_{14}^- \left(\begin{array}{c} \alpha, \alpha^* \\ \gamma, \gamma^* \end{array} \right) &= 6d, 7s, 8s, 9s, 3e \\
C_{14}^- &= 4d, 1s, 2s, 3s, 3e \\
K_{15}^- \left(\begin{array}{c} \beta, \beta^* \\ \gamma, \gamma^* \end{array} \right) &= 6d, 7s, 8s, 9s, 2e \\
C_{15}^- &= 5d, 4s, 5s, 6s, 1e \\
K_\lambda^\pm &= C_\lambda^\mp, 1 \leq \lambda \leq 15
\end{aligned}$$

(71a)

$$\begin{aligned}
Q_1(\text{I}h \rightarrow \text{I}a) &= Q_1 \left((K_j^{++})_1^{18} \rightarrow K_{123}^{--} (123)d, s, e \right), \\
Q_2^+(\text{I}h \rightarrow \text{II}a) &= Q_2 \left((K_j^{++})_4^{18} \rightarrow (K_i^- C_i^-)_1^{15} \right), \\
Q_2^-(\text{II}h \rightarrow \text{I}a) &= Q_2 \left((K_i^+, C_i^+)_1^{15} \rightarrow (123)d, s, e \right), \\
Q_3(\text{III}h \rightarrow \text{II}a) &= Q_3 (K_j^+, C_j^+ \rightarrow C_j^- K_j^-)_1^{15}, \\
Q_4^+(\text{I}h \rightarrow \text{III}h) &= Q_4 \left((K_j^{++})_4^{18} \rightarrow (K_i^+, C_i^+)_1^{15} \right), \\
Q_4^-(\text{II}a \rightarrow \text{I}a) &= Q_4 \left((K_j^-, C_j^-)_1^{15} \rightarrow (123)d, s, e \right)
\end{aligned}$$

(72)

$$1 \leq k \leq k_{\text{max}} < \infty$$

(73)

$${}^2\bar{\kappa}_{(\lambda)\pm} \sim f_{\lambda}^{\pm 1} = f^{\pm 1}(\mu_{\lambda}), \quad f_{\kappa} f_{\lambda} \sim \left| (e^{\lambda_1 \mu} - E)^{\frac{a}{2}} \delta_{kl} \right|_q$$

$$\frac{f_{\kappa}^2 + f_{\lambda}^2}{f_{\kappa} f_{\lambda}} = \beta_{\kappa \lambda} = \text{const}, \quad \beta_{\kappa \lambda} \neq 2, \quad \beta_{\lambda \lambda} = 2,$$

$$\mu = \mu_{\kappa} + \mu_{\lambda}$$

(74)

$$\ddot{x}^i \binom{\mu v}{\kappa \lambda} = - \left[\begin{array}{c} i \\ kl \end{array} \right] \binom{\mu v}{\kappa \lambda} \dot{x}^k \dot{x}^l,$$

$$\delta S^{n_2 4} \bar{\varrho}^{(\mu v)}_{(\kappa \lambda)}; n \partial n = {}^4 \bar{0}$$

(74a)

$$sp()_{(-)}^{(6,6)}; \left(() - \frac{1}{2} {}^2 \bar{\gamma} sp() \right); sp^4 \bar{\varrho} = \bar{0}$$

(75)

$$N_+ \left(\left[\begin{array}{c} \alpha \\ \beta \end{array} \right], \left[\begin{array}{c} \beta \\ \alpha \end{array} \right] \right) (\alpha, \beta) N_- \left(\left[\begin{array}{c} \alpha \\ \beta \end{array} \right], \left\{ \begin{array}{c} \beta \\ \alpha \end{array} \right\} \right), \quad \alpha \hat{=} (\mu \nu), \beta \hat{=} (\kappa \lambda)$$

(76)

$$q_1 = Q_1, \quad q_{2,3} = Q_2^{\pm}, \quad q_4 = Q_3, \quad q_{5,6} = Q_4^{\pm},$$

$$F_v = \binom{\alpha \beta}{\kappa \lambda} F(q_j)_v \hat{=} N_+(\alpha \beta, \kappa \lambda) N_-, \quad 1 \leq \nu \leq 6,$$

$$AF_p BF_r C = AF_v C, \quad p + r = v, \quad A \neq C,$$

$$\left(() - \frac{1}{2} {}^2 \bar{\gamma} sp() \right); sp^4 \bar{\varrho} = \sum_{k=1}^{L>1} \left(() - \frac{1}{2} {}^2 \bar{\gamma}_k sp() \right); sp^4 \bar{\varrho}_k,$$

$$A = C, \quad \left(() - \frac{1}{2} {}^2 \bar{\gamma} sp() \right); sp^4 \bar{\varrho} = \text{const} ({}_4)$$

(76a)

$$F_v \equiv \binom{\alpha \beta}{\kappa \lambda} F(q_j)_v \omega$$

(76b)

$$q_j \hat{=}(j), \quad F_v \hat{=}(j) \omega$$

(77)

$$\bar{\lambda} \perp \bar{Y}(q), \quad \bar{\lambda}_{(\mu \nu)}^{(\kappa \lambda)} \perp \bar{Y}(z), \quad \bar{Y}(q) = \sum_{i=1}^q \dot{\bar{x}}_i,$$

$$z = z(\kappa \lambda \mu \nu) \leq q \leq 6, \quad \bar{Y}(6) = \bar{v} + i \bar{w},$$

$$w^2 = c^2 + \varepsilon^2 + \dot{\eta}^2,$$

(78)

$$w_f = w \geq c, \quad m \lambda = \mu \lambda_o = \text{const}, \quad \frac{\lambda_{(ab)}}{\lambda_o} \sim E_{(ab)}^{-1}$$

$$2n \left(\frac{\lambda_{(cd)}}{\lambda_o} \right)^4 = (2n-1)^2 (1 + 4(q/\pi)^4)$$

(79)

$$\left[\begin{array}{c} i \\ kl \end{array} \right]_{(3)} + Q_{(3)m}^i \left[\begin{array}{c} m \\ k \end{array} \right]_{(3)} = \alpha_{kl}^i e^{\lambda_{kl} \mu} \left[\frac{1}{2} (E + \cos K) (E + \right.$$

$$\left. + (e^{\lambda_{kl} \mu} - \cos K)^2 \sin^{-2} K \right]^{-a/(2\lambda_{kl})}, \quad K = \lambda_{kl} T,$$

$$T = \sum_{k>3}^q \sqrt{\tau} ({}_k), \quad \mu = \sum_{k=1}^3 \alpha_k ({}_k),$$

$$a = \vec{c}_{(3)} \vec{C}^{-1} = a^*$$

(79a)

$$\left(\left[\begin{array}{c} i \\ kl \end{array} \right]_{(3)} + Q_{(3)m}^i \left[\begin{array}{c} m \\ kl \end{array} \right]_{(3)} \right)_{\max} = \alpha_{kl}^i e^{\lambda_{kl}\mu} (1/2 (E + e^{2\lambda_{kl}\mu}))^{-a/(2\lambda_{kl})},$$

$$K; n = \pm \frac{\pi}{2} (2N + 1)$$

(79b)

$$[3]_{kl}^i; n \rightarrow \psi(r) \sim e^{-\alpha r}, \quad \alpha > 0$$

(79c)

$$r = r(v), \quad \partial r \rightarrow \beta = \text{const} > 0$$

(80)

$$\lambda = \underline{\alpha}\lambda, \quad P_{(\mu\nu)}^{(\kappa\lambda)} = \underline{P}, \quad A(1) = [E - \exp\{-\lambda(iT +$$

$$+ i\delta_{xb}Z + \delta_{xc}\mu)\}]^{-\alpha} (E - \delta_{xb}e^{-i\lambda Z} - \delta_{xc}e^{-\lambda\mu})^{\alpha},$$

$$B(2) = [E - \exp\{-i\lambda(Z + T)\}]^{-\alpha} (E - e^{i\lambda Z})^{\alpha},$$

$$C(1, 2) = [E - \exp\{-\lambda(iT + iZ + \delta_{xd}\mu)\}]^{-\alpha} (E - \delta_{xd}e^{-\lambda\mu})^{\alpha},$$

$$D(1, 3) = [E - \exp\{-\lambda(\mu + iT + i\delta_{xd}Z)\}]^{-\alpha} (E - \delta_{xd}e^{-i\lambda Z})^{\alpha},$$

$$E(2, 3) = [E - \exp\{-\lambda(\mu + iZ + iT)\}]^{-\alpha} (E - e^{-i\lambda T})^{\alpha},$$

$$F(1, 2, 3) = [E - \exp\{-\lambda(\mu + iT + iZ)\}]^{-\alpha},$$

$$\mu = \sqrt{\tau} \sum_{k=1}^3 ()_k, \quad Z = \sqrt{\tau} ()_4,$$

$$T = \sqrt{\tau} ((5 + ()_6), \quad x \hat{=} (a, b, c, d)$$

(80a)

$$\hat{\gamma}_{(x)} = ({}^2\bar{\gamma}_{(\mu\nu)})_3, \quad n({}^2\bar{\gamma}_{(\mu\nu)} \neq {}^2\bar{E}) \leq 9,$$

$$Z = n(n-1)$$

(81)

$$(A \dots E)_{\underline{\alpha}=1} \equiv \left[\begin{array}{c} \alpha\beta \\ \alpha\beta \end{array} \right]$$

(82)

$$(A \dots F) \hat{=} -(1 \dots 6), \quad (A \dots E)_{\underline{\alpha}=1} \hat{=} +(1 \dots 5),$$

$$[3] \hat{=} (+7)$$

(82a)

$$p_k = (\pm k), \quad 1 \leq k \leq 5$$

(83)

$$E_N^{(\pm)} = 2^{N-2} \cdot (N!)^2, \quad Z_N = 2^{N-1} \cdot N!$$

$$S_N = N!$$

(83a)

$$N_x^{(\pm)} = 2^y \cdot \prod_{k=1}^n (N_k!)^2, \quad y = \sum_{k=1}^n N_k - 2n$$

$$n = n(x) \leq 6$$

(83b)

$$Q_x^{(k)} = n_x^{(k)} \cdot (\pm k)_x \sim \bar{\lambda}, \quad n_x^{(k)} \geq 1$$

(84)

$$-(\mu x_\mu) - \hat{=} {}^2\bar{\kappa}_{(\mu)}, \quad (\pm p) - (\mu x_\mu) - (\pm q)$$

$$(\pm p) - (\mu x_\mu) - (\mp q), \quad (\pm p) - (3x_3) - (+7)$$

$$Z((-k)) = Z_k, \quad Z(Z_p, Z_q) \hat{=} \frac{1}{2} (Z_p + Z_q - |Z_p - Z_q|),$$

$$0 < x_\mu \leq Z$$

(84a)

$$Z_p \neq Z_q, \quad x_\mu < Z$$

(85)

$$\begin{aligned} (-p) &\equiv F_p(\mu, N^\pm) = N^+(p) \binom{\kappa\lambda}{v} N^-(p) \\ (-q) &\equiv F_q(\mu, N^\pm) = N^+(q) \binom{\kappa'\lambda'}{v'\mu} N^-(q) \\ (\kappa\lambda v) &\neq (\kappa'\lambda'v'), \quad F_p(N^\pm(p), \mu) = F_q(N^\pm(q), \mu), \\ \mu &\stackrel{\wedge}{=} {}^2\bar{\kappa}_{(\mu)} \end{aligned}$$

(85a)

$$\begin{aligned} \bar{F}_{pq}(\mu) \perp \bar{\sigma}_{pq} &= \omega_{pq} \bar{s}_{pq}, \quad \bar{s}_{pq} = \bar{s}_{pq}^{(o)} \cdot s_{pq}, \\ \left| \bar{s}_{pq}^{(o)} \right| &= 1, \quad \bar{s}_p \parallel \bar{s}_q, \\ \varphi &= \pi/2 \end{aligned}$$

(85b)

$$\bar{s}_p^{(o)} = \pm \bar{s}_q^{(o)}$$

(86)

$$\begin{aligned} \bar{S}[(\pm p)_x - (\mu) - (\pm q)_x] &= {}^p_x(\mu)_\pm^q \\ \cos[(\mu)_+, (\mu)_-] &= -1 \end{aligned}$$

(86a)

$$(\pm p)_{b,c} - (\mu V_\mu) - (\pm q)_{b,c}, \quad V_\mu^x + V_\mu^y = V_\mu$$

(86b)

$$\begin{aligned} (\pm p) - (\mu) - (\pm q), \quad (\pm p) - (\mu V_\mu^x) - (\pm k) - (\mu V_\mu^y) - (\pm q), \\ {}^k_x(\mu)_\pm^{p,q} (V_\mu^x, V_\mu^y), \quad I(k) > \frac{1}{2}[I(p) + I(q)] \end{aligned}$$

(87)

$${}^p_x(\mu)_\pm^q \sim S_{\bar{w}}^{\bar{\xi}_\mu} \sum_{\varepsilon} s p \Lambda_{(\kappa\lambda)}^{(\mu v)}; \left[\frac{\widehat{\mu v}}{-\frac{1}{\kappa\lambda}} \right]_+; () \prod_{k=1}^3 \partial()_k$$

(87a)

$${}^p_x(\mu)^q = \pm \bar{s}_\mu \frac{\hbar}{2} \underline{m}_x^{(pq)}, \quad {}^p_x(\mu)^q = \bar{s}_\mu |^p_x(\mu)^q |$$

(88)

$$\bar{s}_x = \sum_{\mu,p,q} {}^p_x(\mu)^q = \pm \bar{s}_0 \hbar m_x / 2$$

(88a)

$$\bar{s}(x) = \pm \hbar \bar{\sigma}(x), \quad 2\bar{\sigma}(x) = \bar{s}_0 m_x; v, \quad \bar{\sigma} \neq \bar{\sigma}^*$$

(89)

$$\begin{aligned} \bar{\sigma}(x) &= i(\bar{e}_t s + \bar{e}_r J P_a), \quad \bar{e}_t \perp \bar{e}_r, \quad 2s = P \geq 0, \\ 2J = Q \geq 0, \quad P_a &= \alpha_p (-1)^J, \quad \alpha_p = \pm 1 \end{aligned}$$

(89a)

$$\sigma_t = i s, \quad \sigma_r = i J P_a$$

(90)

$$s_v = s + 1 - v, \quad 1 \leq v \leq I = P + 1$$

(90a)

$$0 \leq P \leq G = k + 1$$

(91)

$$\begin{aligned} Q = 2n, \quad \sigma_r &= \pm i J_T, \quad J_T = n, \\ Q = 2n + 1, \quad \sigma_r &= \pm J_S, \quad 2J_S = 2n + 1 \\ n &\geq 0 \end{aligned}$$

(91a)

$$a_g \hat{=} a(R_4), \quad P(a_g) = P(b) = 0, \quad J_T(a_g) = 2, \\ J_T(b) = 1$$

(92)

$$K(R_4^\pm) = K^\pm, \quad \cos(\vec{Y}^+, \vec{Y}^-) = -1, \quad \bar{\lambda}^\pm \perp \vec{Y}^\pm, \\ \bar{\lambda}^+ \parallel \bar{\lambda}^-, \quad a^+(\omega) = a^-(\omega), \quad b^+ = b^-, \\ m^\pm > 0$$

(92a)

$$I_+(X) \equiv I_-(\bar{X}) = \bar{I}$$

(93)

$$\varepsilon(R_4^\pm) = \cos(\vec{S}(\vec{\lambda}), \vec{x}_4) = \pm 1, \quad C_\pm(R_4^\pm) = -C_\mp \\ \varepsilon C_\pm = C > 0$$

(94)

$$\bar{\lambda}_X - [X, Y] - \bar{\lambda}_Y, \quad J_X > 0, \quad J_Y > 0 \\ \cos(\bar{s}_X, \bar{s}_Y) \approx \pm 1$$

(95)

$$Z(x) = n(n-1), \quad n = n(\hat{\gamma}_{(x)})$$

(95a)

$$(+p)_X - [\hat{\mu}] - (+q)_Y \rightarrow (\pm p)_X - \langle \mu \rangle - (\pm q)_Y \rightarrow \\ \rightarrow (\pm p)_X - \langle \mu \rangle - (\pm q)_Y \rightarrow (\pm p)_X - \langle \mu \rangle - (\pm q)_Y + \\ + (\pm p)_X - \langle \mu \rangle - (\pm q)_Y \hat{=} (\pm p)_X - \langle \hat{\mu} \rangle - (\pm q)_Y$$

(96)

$$\bar{\bar{\lambda}} = \underline{N} \bar{\lambda}_0 \perp \vec{Y}, \quad \underline{N} \geq 1, \quad \underline{N} = N_{(\mu\nu)}^{(\kappa\lambda)}$$

(96a)

$$\underline{\varrho}_n = \underline{n}\varepsilon, \quad \bar{\lambda}_0 = \bar{\lambda}_0(k), \quad |\bar{\lambda}|\varepsilon = 1, \quad \underline{n} \geq 1$$

(96b)

$$m_L c s_0 = 4 \sqrt[4]{\pi} \sqrt[3]{3\pi s_0 \gamma \hbar} \sqrt{\frac{c\hbar}{3\gamma}}, \\ m_e \eta \sqrt[3]{\eta} = m_L \left(1 - \frac{2}{3} f \xi \eta^2 (1 - \sqrt{\eta}) Y_2 \right), \quad 2\xi = 1 + \sqrt{5}, \\ m_B \eta = m_e, \quad f = \beta + \alpha / (8\pi e), \quad \beta \approx 1$$

(97)

$$\mu_{\pm} = 4\mu\alpha_{\pm}, \quad \mu_S = \left(1 - \frac{\alpha_-}{\alpha_+}\right)\mu_+ \\ c\mu s_0 = \sqrt[4]{\pi} \sqrt[3]{3\pi s_0 \gamma \hbar} \cdot \sqrt{\frac{c\hbar}{3\gamma}} \\ \eta^2(1 + \alpha_+) = \eta(1 + \alpha_-) = \frac{1}{\sqrt[3]{\eta}} \left(1 - \frac{2}{3} \xi \eta^2 (1 - \sqrt{\eta}) Y_2\right) \\ s_0 = 1[\text{ m}]$$

(97a)

$${}^2\bar{P}_{(\kappa\lambda)}^{(\mu\nu)}; N = {}^2\bar{A}_{(\kappa\lambda)}^{(\mu\nu)} + i^2\bar{F}_{(\kappa\lambda)}^{(\mu\nu)}; N, \quad {}^2\bar{A}_{(\kappa\lambda)}^{(\mu\nu)} = \text{const}(R_6)$$

(98)

$$e_\varrho = \varepsilon_\pm \sqrt{\eta_{qk}}, \quad 2e_\omega = \varepsilon_\pm (1 + \sqrt{\eta_{qk}}) \\ e_\delta = \varepsilon_\pm (1 - \sqrt{\eta_{qk}}), \quad e_C = \varepsilon_\pm \sqrt{\vartheta_{qk}/8} \\ \vartheta_{qk} = 5\eta_{qk} + 2\sqrt{\eta_{qk}} + 1, \quad \eta_{qk} \sqrt[4]{\pi^4 + q^4(4+k)} = \pi$$

(98a)

$$k_{\max} = 2, \quad q_{\max} = 3$$

(98b)

$$\begin{aligned} N_{(j)}(t) &= n_j(t) + Q_j, \quad (j \leq 4) \hat{=} (n, m, p, \sigma), \\ Q_n &= 3 \cdot 2^{s-2}, \quad Q_m = 2^s - 1, \quad Q_p = 2^s + 2(-1)^k, \\ Q_\sigma &= 2^{s-1} - 1, \quad s = k^2 + 1 = \text{const}(t) \end{aligned}$$

(98c)

$$\begin{aligned} 2\alpha_1 &= 1 + \sqrt{\eta_{qk}}, \quad \alpha_2 \eta_{qk} = 1, \quad k\alpha_3 = e^{k-1} - kqF, \\ \alpha_4 &= 1, \quad F = H + G, \quad 3H = \alpha(1 + \sqrt{\eta_{qk}}) \left(\frac{\xi}{\eta_{qk}^2} \right)^{2k+1} \eta_{qk}^3, \\ e\eta_{qk}G &= \eta_{11} (2\xi\eta_{qk})^k \left(1 - \frac{\sqrt{\eta_{qk}}}{1 + \sqrt{\eta_{qk}}} \right)^2 \end{aligned}$$

(98d)

$$M(c, d) = T; m = \mu_+ \left[\sum_{j=1}^4 \alpha_j G_j + \left(1 - \frac{\alpha_-}{\alpha_+} \right) F_S + q \frac{\alpha_-}{\alpha_+} \right]$$

(98e)

$$\begin{aligned} \eta_j N_{(j)} &= G_j, \quad 4G_1 = N_{(1)}^2 (1 + N_{(1)})^2, \\ 6G_2 &= N_{(2)} (2N_{(2)}^2 + 3N_{(2)} + 1), \quad 2G_3 = N_{(3)} (1 + N_{(3)}) \\ G_4 &= N_{(4)}, \quad \eta_1 \geq \eta_2 \geq \eta_3 \geq \eta_4 = 1 \end{aligned}$$

(99)

$$\begin{aligned} k_{\max} &= 2, \quad 0 \leq P \leq k+1, \quad Q(P) = k-1, \\ Q(P_{\pm}) &= 2k-1, \quad P_+ = 2k-1, \quad P_- = 2-k \end{aligned}$$

(99a)

$$\kappa(\lambda) = (1 - \delta_{1\lambda}) \delta_{1P}, \quad 1 \leq \lambda \leq \lambda_k = 4 - k$$

(100)

$$\begin{aligned} Ck(1 + \kappa) &= 2(P\varepsilon_P + Q\varepsilon_Q)(k-1 + \kappa), \quad \varepsilon_{P,Q} = \varepsilon \cos \alpha_{P,Q}, \\ \alpha_P &= \pi Q \left(\kappa + \binom{P}{2} \right), \quad \alpha_Q = \pi Q \left(Q(k-1) + \binom{P}{2} \right) \end{aligned}$$

(100a)

$$\begin{aligned} 2q_x &= (P - 2x)(1 - \kappa Q(2 - k)) + \\ &+ \varepsilon(k - 1 - (1 + \kappa)Q(2 - k)) + C, \quad 0 \leq x \leq P, \\ |q_x| &= q \end{aligned}$$

(100b)

$$q_x(R_4^+) + q_x(R_4^-) = C(R_4^+) + C(R_4^-) = 0$$

(101)

$$(v)(kPQ\kappa)\varepsilon C(\varepsilon q_x)_0^P, \quad 1 \leq v \leq 12$$

(101b)

$$\begin{aligned} (1) \hat{=} (\eta), \quad (2) \hat{=} (e_0, e^-), \quad (3) \hat{=} (\mu^-), \\ (4) \hat{=} (K^+, K^0), \quad (5) \hat{=} (\pi^+, \pi^0, \pi^-), \quad (6) \hat{=} (\Lambda), \\ (7) \hat{=} (\Omega^-), \quad (8) \hat{=} (p, n), \quad (9) \hat{=} (\Xi^0, \Xi^-), \\ (10) \hat{=} (\Sigma^+, \Sigma^0, \Sigma^-), \quad (11) \hat{=} (o^+, o^0, o^-, o^-), \\ (12) \hat{=} (\Delta^{++}, \Delta^+, \Delta^0, \Delta^-), \quad k = B + 1 \end{aligned}$$

(101c)

$$(v)(\varepsilon)(\varepsilon B, \varepsilon P, \varepsilon Q, \varepsilon \kappa) C(q_x)_0^P, \quad (v)(\varepsilon = +1) \equiv (v),$$

$$(v)(\varepsilon = -1) \equiv (\bar{v}), \quad (5) \hat{=} (\pi^\pm, \pi^0) \equiv (\bar{5})$$

(102)

$$\hat{I} \equiv kP \left(\begin{array}{c} Q, \kappa \\ C, q_x \end{array} \right]_{\varepsilon}$$

(103)

$$n_j = 0, \quad 1 \leq \lambda \leq G = k + 1, \quad z_\lambda(b_\lambda) \geq 16$$

(103a)

$$\begin{aligned} G(1) &= 2, & b_1 &\equiv Z(n), & b_2 &\equiv Z(m) + Z(p) + Z(\sigma), \\ G(2) &= 3, & b_1 &\equiv Z(n), & b_2 &\equiv Z(m), \\ b_3 &\equiv Z(p) + Z(\sigma) \end{aligned}$$

(104)

$$4\pi^2 C_{\pm} = \pm \sqrt{2v \frac{\hbar}{R_-}}, \quad e_{\pm} = 3C_{\pm}$$

(105)

$$\begin{aligned} (2\pi)^5 \alpha \sqrt{1 - \alpha^2} &= 9\vartheta \left(1 - A_1 A_2 Y_3\right), \quad (1 + \sqrt{\eta_{1k}}) A_k = \\ &= (1 - \sqrt{\eta_{1k}}) \sqrt{\eta_{1k}}, \quad \alpha > 0 \end{aligned}$$

(105a)

$$\alpha_{(+)} = \alpha, \quad \alpha_{(-)} = \beta, \quad \beta \approx 137\alpha$$

(106)

$$\mathrm{rot}\,\underline{B}=2A\sigma(\vec{r}\times\vec{w}),\quad \pi cA=\sqrt{3\pi\gamma\mu_0}=\pi cA'\sqrt{3/2}$$

(106a)

$$\begin{aligned} 2B_H &= B \cos \varphi, & B_V &= B \sin \varphi, & B &= 2AD\mu S(t), \\ D &= \sigma R^2 w, & \mu(x_1, x_2, x_3) &> 0, & 0 &\leq S(t) \leq +1 \end{aligned}$$

(106b)

$$D>\frac{\Delta}{2\mu A}$$

(107)

$$\partial_j G_j > G_{j+1}, \quad \partial_j G_j \geq \partial_{j+1} G_{j+1}$$

(107a)

$$\begin{aligned} j > 1, \quad \beta_j &= {}_{j-1}G_{j-1} - G_j \geq 1, \quad \beta_j = 0, \\ G_j &= 0, \quad n_{j-1} \rightarrow 1 + n_{j-1}, \quad j = 1, \quad \partial_1 G_1 > 0, \\ -Q_j &\leq n_j \leq L_j < \infty \end{aligned}$$

(107b)

$$\beta_4=\alpha_3N_{(3)}-N_{(4)}>0, \quad \mu_+c^2\approx 9,28717[\mathrm{KeV}]$$

(108)

$$\begin{aligned} \alpha_1 N_{(1)}^3 + \alpha_2 N_{(2)}^2 + \alpha_3 N_{(3)} + \exp\left(-\frac{2k-1}{3Q_4} N_{(4)}\right) &= \\ = W(vx)(1+f(N)), \quad W(vx) &= g(k,q)w(vx), \\ g(k,q) &= \alpha_1 Q_1^3 + \alpha_2 Q_2^2 + \alpha_3 Q_3 + \exp\left(-\frac{1}{3}(2k-1)\right) \end{aligned}$$

(108a)

$$\begin{aligned} f(N \geq 0) &\equiv 0, \quad \partial_N f > 0, \quad \partial_N N = \partial N = 1, \\ 0 &\leq N \leq L_N < \infty, \quad w(vx) = 1 + (2-k)\underline{w}_1 + (k-1)\underline{w}_2, \\ w &= w^*, \quad \underline{w}_k(n_j = 0) = 0 \end{aligned}$$

(109)

$$w(vx) = w_1^{2-k} + w_2^{k-1}$$

(109a)

$$\begin{aligned} w_1 + 1 - k &= (1 - Q) \left[A_{11} - P \left(A_{12} + \frac{\kappa q}{\eta_{qk}} A_{13} \right) - \binom{P}{2} (A_{14} - \right. \\ &\quad \left. - \frac{q}{\eta_{qk}} A_{15}) \right] + \kappa Q \eta_{qk} A_{16}, \quad w_2 - 2 + k = (q - 1) A_{21} + \\ &+ (1 - P) A_{22} + \binom{P}{2} (A_{23} - q_x \eta_{qk} (1 + A_{24} (1 + q_x))^{-1} A_{25}) + \\ &+ \kappa (A_{26} + q \eta_{qk}^2 A_{31}) + \binom{P}{3} [q^3 (3 - q)^{-1} (\varepsilon q_x - (-1)^q) A_{33} + \\ &+ \frac{e(P - Q) \eta_{qk}}{(8 - q(q - 1) A_{66}) \eta^2} \left(1 - \frac{q}{\eta_{qk}} (2 - \right. \\ &\quad \left. - q) A_{34}^{1 - \varepsilon q_x} A_{35}) \eta^{(q-1)q/4} - A_{36}] + \binom{Q}{3} \eta_{qk} A_{32} \end{aligned}$$

(109b)

$$\begin{aligned} A_{21} &= 2 \left(\frac{e\alpha}{2\eta} \right)^2 \left(1 - \frac{\alpha}{2\xi^2} \right) Y_{10}, \\ 12A_{22} &= \xi \left(1 - \xi \left(\frac{\alpha\xi}{\eta^2} \right)^2 \right) Y_{11}, \\ eA_{23} &= (\eta^2 + 6\xi\alpha^2) Y_{12}, \quad 3\eta A_{24} = 2\xi^2 Y_{13}, \\ 3A_{25} &= \pi\xi(\pi e)^2 (1 - \beta^2) \eta \sqrt{\eta} Y_{14}, \\ e\xi^2 A_{26} &= 2 \left(1 - \frac{\pi}{2} (e\xi\alpha)^2 \sqrt{\eta} \right) Y_{15}, \\ A_{31} &= (\pi e\alpha)^2 (1 - (\pi e)^2 (1 - \beta^2)) Y_{16}, \\ 6\beta A_{32} &= \xi^2 \left(1 + \frac{2\alpha}{3\pi} Y_{17} \right), \\ A_{33} &= (\pi e\xi)^2 (1 - 2\pi(e\xi)^2 (1 - \beta^2)) Y_{18}, \\ A_{34} &= \eta \sqrt{2\pi\eta} Y_{19}, \quad e\xi^2 A_{35} = 3\alpha Y_{20}, \\ A_{36} &= (1 - \pi e(\xi e)^2 (1 - \beta^2))^{-1} Y_{21}, \quad A_{66} = \xi \eta Y_{22} \end{aligned}$$

(110)

$$f(N) = (1 - Q(1 - \kappa)(2 - k)Y_{23}) \left(\frac{aN}{N + 2} + b\sqrt{N(N - 2)} - ib\delta_{1N} \right)$$

(110a)

$$\begin{aligned} ka(vx) &= A_{41} (1 + a_n a_q), \quad a_n = P A_{42} [1 - \kappa A_{43} (1 + \\ &\quad + A_{44} (-\alpha)^{2-k} A_{45}^{k-1}) (1 - \kappa Q A_{46} (2 - k)) - A_{51} (k - 1)(1 - \kappa)], \\ a_q &= 1 - q A_{52} (1 - 2\kappa A_{53}^k) \left(1 + \frac{q\kappa}{6} (3 - \varepsilon q_x) (k - 1)(1 - \kappa) \right) \end{aligned}$$

(110b)

$$\begin{aligned} k^P (1 + P + Q + \kappa \eta^{2-q}) (1 + R(S_1 + S_2)) b(vx) &= \\ &= A_{54} A_{55}^{k-1} [1 + P A_{56} (1 - \kappa A_{61} A_{62}^{1-k}) (1 + q A_{63} (1 + \\ &\quad + \kappa A_{64}))] \left(1 - k^{-1} (A_{65} (q + k - 1))^{2-k} \binom{P}{2} \left(1 - \binom{P}{3} \right) \right) \\ S_1 &= \binom{P}{2} (2 - k)(1 - q), \quad S_2 = P \left(1 - \binom{P}{2} \right) (1 - \\ &\quad - \binom{P}{3}) (k - 1)(1 - \kappa) \end{aligned}$$

(110c)

$$\begin{aligned}
(2\beta - \alpha)A_{41} &= \beta \left(\xi (2 + (\xi\alpha)^2) - 2\beta \right) Y_{24}, \\
2A_{42} &= \pi\xi^2\eta(\beta - 3\alpha)Y_{25}, \quad 2A_{43} = \xi Y_{26}, \\
A_{44} &= 2 \left(\frac{\eta}{\xi} \right)^2 Y_{27}, \quad 6\xi A_{45} = (3\beta - \alpha)Y_{28}, \\
A_{46} &= \left(\frac{\pi e}{\xi\eta} - e\eta^2\alpha/2 \right) Y_{29}, \quad A_{51} = (2\alpha + 1)^2 Y_{30}, \\
\eta^2 A_{52} &= 6\alpha Y_{31}, \quad A_{53} = \left(\frac{\xi}{\eta} \right)^3 Y_{32}, \quad A_{54}\sqrt{2} = \alpha(\beta - \\
&\quad - \alpha)\sqrt{3}Y_{33}, \quad A_{55} = \xi^3 Y_{34}, \quad A_{56} = \left(\frac{\xi}{\eta} \right)^4 Y_{35}, \\
12\beta A_{61} &= \pi\xi(2\beta - \alpha)Y_{36}, \quad 12A_{62} = \pi^2(\beta - 2\alpha)Y_{37}, \\
9A_{63} &= \sqrt{\eta}Y_{38}, \quad 3\eta A_{64} = \pi Y_{39}, \quad 3\xi A_{65} = \pi Y_{40}
\end{aligned}$$

(110d)

$$\hat{A} = (A_{ik})_6 = \hat{A}^* \neq \hat{A}^\times$$

(111)

$$\phi = A_v F_1 F_q F_\kappa / F_2 + B_v (P + Q) + 4q\alpha_- / \alpha_+$$

(111a)

$$\begin{aligned}
F_1 &= P(P+Q)(-1)^{P+Q} (2-k+\eta^3(k-1)), \\
F_2 &= 1 + 4\frac{\xi}{k} \binom{P}{2} (\xi/6)^q Y_{41}, \\
F_q \sqrt{\eta_{qk}} &= \left(3 - \alpha + \frac{\pi}{2} (k-1) 3^{2-q/2} \right) (2\sqrt{\eta_{11}\eta_{qk}} + q\eta^2(k-1)Y_{42}), \\
F_\kappa &= 1 + \frac{2k\kappa}{3\eta^2} \xi (1 + \pi\xi^2(P-Q)(\pi - 5q/4)) Y_{43}
\end{aligned}$$

(111b)

$$\begin{aligned}
\beta\pi A_v &= (1 - \alpha_- / \alpha_+) \left((\pi/3)^2 + (\eta/\eta_{11})^2 \frac{\alpha}{3\xi} \right) \\
\xi^2 B_v &= \alpha (1 - \alpha_- / \alpha_+)^2 Y_{44}
\end{aligned}$$

(112)

$$M(N) = \mu\alpha_+(K + F + H + \phi)$$

(112a)

$$\begin{aligned}
K &= N_1 Q_1^2 (1 + Q_1)^2 + N_2 Q_2 (2Q_2^2 + 3Q_2 + 1) + \\
&\quad + N_3 Q_3 (1 + Q_3) + 4Q_4, \quad F = N_1 n_1^2 (1 + n_1)^2 + \\
&\quad + N_2 n_2 (2n_2^2 + 3n_2 + 1) + N_3 n_3 (1 + n_3) + 4n_4 \\
H &= 2n_1 Q_1 (1 + 3(n_1 + Q_1 + n_1 Q_1) + 2(n_1^2 + Q_1^2)) N_1 + \\
&\quad + 6n_2 Q_2 (1 + n_2 + Q_2) N_2 + 2n_3 Q_3 N_3
\end{aligned}$$

(112b)

$$N_1 = \alpha_1, \quad 3N_2 = 2\alpha_2, \quad N_3 = 2\alpha_3$$

(113)

$$k\mu_+\alpha_1 (L_1 + Q_1)^3 = 2(k+1)(P+1)^{2/k} M_x Y_{45}$$

(113a)

$$\begin{aligned}
\alpha_2 (L_2 + Q_2) &\left(2(L_2 + Q_2)^2 + 3(L_2 + Q_2) + 1 \right) = 6\alpha_1 (L_1 + Q_1)^3, \\
\alpha_3 (L_3 + Q_3) &(1 + L_3 + Q_3) = 2\alpha_2 (L_2 + Q_2)^2, \\
L_4 + Q_4 &= TRC(\alpha_3 (L_3 + Q_3))
\end{aligned}$$

(113b)

$$\alpha_1 (L_1 + Q_1)^3 + \alpha_2 (L_2 + Q_2)^2 + \alpha_3 (L_3 + Q_3) + \\ + \exp \left(-\frac{2k-1}{3Q_4} (L_4 + Q_4) \right) = W(vx) (1 + f(L_N))$$

(113c)

$$M_{\max}(L_j)_1^4 \geq M_L(L_N), \quad E(\beta_{j>1} > 0) = (M_{\max} - M_L) c^2$$

(114)

$$Q(N) = Q + Z(N), \quad Q(0) = Q, \quad Z(0) = 0, \\ q_x(N) \neq q_x(0) = q_x$$

(115)

$$\hat{S} \equiv kP \left(\begin{array}{c|c} n,m & Q\kappa \\ p,\sigma & \left[\begin{array}{c} N \\ C_{q_x} \end{array} \right]_{\varepsilon} \end{array} \right)^N X(vx), \quad (n,m,p,\sigma) = n_1, \dots, n_4$$

(116)

$$\lambda c \left(M\left(N_m \right) - M\left(N_m + 1 \right) \right) = h$$

(117)

$$h\omega_R > (M(N) - M_x) c^2, \quad \omega_R = \omega_x = 1/\theta_x, \\ T > \theta_x > 0$$

(118)

$$\left(\begin{array}{c} -Q_1, -Q_2 \\ -Q_3, -Q_4 \end{array} \middle| \right) \equiv \left(0 \middle|, \quad \hat{S}(v_x) = kP \left(0 \middle| \begin{array}{c} Q \kappa v_x v_x \\ \tilde{C}_0^\varepsilon \end{array} \right)$$

(118a)

$$m(v_x)=\mu\alpha_+\phi_{q=0}\neq 0$$

(118b)

$$\phi_e = 4A_v(3-\alpha)\sqrt{\eta_{11}} + 2\text{ B}_v, \quad \phi_\mu = 4A_v\left(1 + \frac{2\xi}{3v^2}\right)(3- \\ -\alpha)\sqrt{\eta_{11}} + 2B_v, \quad \phi_\pi = 4A_v(3-\alpha)(1+4\xi)^{-1}\sqrt{\eta_{11}} + 2B_v, \\ \phi_R = \phi_P = B_v, \quad \phi_\beta = 2B_v$$

(118c)

$$v_\beta \rightrightarrows v_P + v_R$$

(15a')

$$F(x_1\dots x_6)=N\tau, \quad N>0$$

(M1)

$$y=f(x), \quad \lim_{\Delta F_v \rightarrow \tau} \sum_{v=1}^n \tilde{y}_v \Delta x_v = n\tau, \\ \Delta F_v = \tilde{y}_v \Delta x_v = \int_{x_{v-1}}^{x_v} y dx, \quad \sum_{v=1}^n \int_{v-1}^v = \int_0^n, \\ \int_{x_0}^{x_n} f(x) dx = n\tau, \quad x_n = x(n), \quad y_n = f(n)$$

(M2)

$$\partial \varphi = \varphi(n) - \varphi(n-1), \quad 1 \leq n \leq N$$

(M2a)

$$J\left(n_1,n_2\right)=\overset{n_2}{n_1}\varphi(n)\,\mathfrak{d}\,n,\quad \mathbf{S}\hat{=}\sum,\quad n_1\geqslant 1, \\ n_2>n_1$$

(M3)

$$\partial^k \varphi = \sum_{v=0}^k (-1)^v a_v(k) \varphi(n-v), \quad a_v(k) = \binom{k}{v}$$

(M6a)

$$\mathbf{S} \frac{\varphi}{\psi} \partial n = \frac{u}{v}, \quad \frac{\varphi}{\psi} = \frac{vu - uv}{v(n)v(n-1)}$$

(M7)

$$0 < |\partial_\varepsilon| \ll 1, \quad \partial_\varepsilon \varphi \approx \varphi \partial_\varepsilon \ln \varphi, \quad \partial_\varepsilon e^\varphi \approx e^\varphi \partial_\varepsilon \varphi$$

(M8)

$$f = f(\varphi), \quad \partial_\varphi f \partial \varphi = f(\varphi) - f(\varphi - \partial \varphi)$$

(M9)

$$\begin{aligned} \varphi &= \varphi(n_1 \dots n_L) = \varphi(n_i)_1^L, \quad 1 \leq \kappa_i \leq n_i \leq N_i < \infty, \\ \partial_i \varphi &= \varphi - \varphi(\dots, n_i - 1, \dots), \quad \partial = \sum_{i=1}^L \partial_i, \\ (\partial_i \times \partial_k)_- &= 0 \end{aligned}$$

(M9a)

$$\begin{aligned} \varphi &= \varphi(\psi_1, \dots, \psi_\lambda), \quad \psi_j = \psi_j(n_1 \dots n_L), \\ \partial \varphi &= \sum_{j=1}^\lambda \partial_{\psi_j} \varphi \partial \psi_j, \quad \partial_{\psi_j} \varphi = (\varphi - \varphi(\dots, \psi_j - \partial \psi_j, \dots)) (\partial \psi_j)^{-1} \end{aligned}$$

(M10)

$$\phi = \mathbf{S}_{1+\kappa_1}^{N_1} \dots \mathbf{N}_L \varphi(n_i)_1^L \prod_{k=1}^L \partial n_k, \quad \partial_i n_k = \delta_{ik}$$

(M11)

$$\begin{aligned} Z(i) &= ()_i, \quad \varphi(n_i)_1^L = \phi; n, \quad \phi = \phi(C_k, Z(i))_{i,k=1}^{L,K} \\ C_k; n_i &= f_k(n_i) \end{aligned}$$

(M11a)

$$\begin{aligned} 0; n = 0, \quad E; n = 1, \quad E; () ; n = n, \quad (C_i \times C_k)_\pm \neq 0, \\ C; n = a = \text{const}(n), \quad C = a \frac{()}{()} \end{aligned}$$

(M11b)

$$\bar{Z}(i) = \bar{e}_i()_i, \quad |\bar{e}_i| = 1, \quad (\bar{e}_i \bar{e}_k)_L = \hat{A}(n_i)_1^L$$

(M11c)

$$\begin{aligned} \bar{C}_i &= \bar{e}_i C_i, \quad C_i; n = \varphi_i(n_k)_1^L, \quad {}^m \bar{C} = [m] [\prod_{k=1}^m C_{i_k}] \\ {}^m \bar{T} &= {}^m \bar{C}; n, \quad 0 \leq m \leq L \end{aligned}$$

(M12)

$$\begin{aligned} {}^2 \bar{C} &= [C_i; C_k]_L, \quad {}^2 \bar{C} = {}^2 \bar{C}_+ + {}^2 \bar{C}_-, \\ {}^2 \bar{C}_\pm &= \frac{1}{2} [(C_i \times C_k)_\pm]_L, \\ {}^2 \bar{C}_+ &= {}^2 \bar{C}_+^\times, \quad {}^2 \bar{C}_- = -{}^2 \bar{C}_-^\times \end{aligned}$$

(M12a)

$$\begin{aligned} sp^2 \bar{C} &= \sum_{i=1}^L C_i^2, \quad sp_{i=k} {}^m \bar{C}_{+(i,k)} = {}^{m-2} \bar{C}, \\ sp_{i=k} {}^m \bar{C}_{-(i,k)} &= {}^{m-2} \bar{0} \end{aligned}$$

(M12b)

$$\begin{aligned} \bar{D}; {}^m \bar{C} &= {}^{m+1} \bar{W}, \quad sp \bar{D}; {}^m \bar{C} = {}^{m-1} \bar{W}, \quad D; {}^m \bar{C} = {}^m \bar{W}, \\ \bar{\partial} &= \sum_{i=1}^L \bar{e}_i \partial_i, \quad -\varphi = \text{GRAD}_L \varphi, \\ \bar{\partial}; {}^m \bar{C} &= \widehat{\text{DIV}}_L {}^m \bar{C}, \quad sp \bar{\partial}; {}^m \bar{C} = \overline{\text{DIV}}_L {}^m \bar{C}, \\ \bar{\partial}; {}^m \bar{C} - (\bar{\partial}; {}^m \bar{C})^\times &= \text{ROT}_L {}^m \bar{C} \end{aligned}$$

(M23)

$$N = \bar{K} \bar{\partial} \bar{n}, \quad \bar{n} = \sum_{k=1}^N \bar{e}_k Z(k); n$$

(M23a)

$${}^2\bar{K} = {}^2\bar{\kappa}; n$$

(M24)

$${}^2\bar{\gamma}_{(ab)} = \text{sp} \left({}^2\bar{a} \times {}^2\bar{b} \right), \quad 2[pkl]_{(ab)} = \frac{1}{\alpha_k} \partial_k \gamma_{(ab)pl} + \\ + \frac{1}{\alpha_l} \partial_l \gamma_{(ab)kp} - \frac{1}{\alpha_p} \partial_p \gamma_{(ab)kl}, \quad [3] \left[\left[\begin{smallmatrix} pkl \\ (ab) \end{smallmatrix} \right]_N \right] = [ab$$

(M24a)

$${}^2\bar{\gamma}_{(cd)} = \text{sp} \left({}^2\bar{c} \times {}^2\bar{d} \right), \quad \gamma^i(c d)[pkl]_{(ab)} = \left[\begin{smallmatrix} i \\ kl \end{smallmatrix} \right]_{(ab)}^{(cd)}, \\ [3] \left[\left[\begin{smallmatrix} i & \\ k & l \end{smallmatrix} \right] \left(\begin{smallmatrix} cd \\ ab+ \end{smallmatrix} \right) \right]_N = \left[\begin{smallmatrix} \widehat{cd} \\ -a \end{smallmatrix} \right]$$

(M25)

$$\partial_p^2 n^i + \frac{\alpha_k \alpha_l}{\alpha_i} \partial_p n^k \partial_p n^l \left[\begin{smallmatrix} i \\ k \\ l \end{smallmatrix} \right]_{(ab)}^{(cd)}; n = 0, \quad n^i = n^i(p), \\ \partial_p^2 \xi^i = 0, \quad \left[\begin{smallmatrix} \widehat{cd} \\ a \end{smallmatrix} \right]_{(\xi)} = \hat{0}, \quad \left[\begin{smallmatrix} \widehat{cd} \\ ab \end{smallmatrix} \right] \neq \hat{0}$$

(M25a)

$$\partial_p^2 x^i + \partial_p x^{\not k} \partial_p x' \left[\begin{smallmatrix} i \\ kl \end{smallmatrix} \right]_{(ab)(C')}^{(cd)}; n = 0$$

(M25b)

$$\partial_{x' \underline{m}' \underline{u}}^2 x'^i + \left[\begin{smallmatrix} i \\ k & l \end{smallmatrix} \right]_{(ab)(C'')}^{(cd)}; n \partial_{x' \underline{m}} x'' \underline{k} \partial_{x' \underline{u}} x'' \underline{l} = \\ = [m^p \mu]_{(ab)(C')}^{(cd)}; n \partial_{x' \underline{p}} x''^i$$

(M28a)

$$\lim_{2\hat{\gamma} \rightarrow 2 \cdot E} (\kappa)_{(\pm)}^{(s_1)(s_2)} = \widehat{\text{DIV}}_{(x)}$$

(M28b)

$$(\kappa)_1 = \frac{1}{\alpha_l} \partial_l - \left[\begin{smallmatrix} s \\ ls \end{smallmatrix} \right]_{(\kappa)+}^{(\kappa)}; n, \quad \lim_{2\hat{\gamma} \rightarrow 2 \cdot E} (\kappa) = \text{GRAD}_{(x)}$$

(M29)

$$\frac{1}{\alpha_m} \partial_m \frac{1}{p} (\kappa)_l; p - \frac{1}{\alpha_l} \partial_l \frac{1}{p} (\kappa)_m; p = \\ = \frac{1}{\alpha_l} \partial_l \left[\begin{smallmatrix} s & s \\ m & \end{smallmatrix} \right]_+; n - \frac{1}{\alpha_m} \partial_m \left[\begin{smallmatrix} s & \\ l & s \end{smallmatrix} \right]_+; n, \quad \left[\begin{smallmatrix} i \\ k \end{smallmatrix} \right]_{(\kappa)}^{(\kappa)} = \left[\begin{smallmatrix} i & \\ k & l \end{smallmatrix} \right]$$

(M29a)

$$sp \left((\kappa)_{(+)}^{(1)} + (\kappa)_{(+)}^{(2)} \right); \bar{A} = 2 \text{DIV}_{(x)} \bar{A}, \\ sp \left((\kappa)_{(+)}^{(1)} - (\kappa)_{(+)}^{(2)} \right); \bar{A} = 2 \underline{A}^k \left[\begin{smallmatrix} sk \end{smallmatrix} \right] (\kappa); n, \quad \bar{A} = p \bar{A}, \\ (\kappa)_{(+k)}^{(1,2)} \cdot \underline{y}^{ik} = \frac{1}{\alpha_k} \partial_k \underline{\gamma}^{ik} - \left[\begin{smallmatrix} s \\ sk \end{smallmatrix} \right]_{(\kappa)-}^{(\kappa)}; n \cdot \underline{y}_{+}^{ik}, \\ \gamma_{ik} (\kappa)_{(+l)}^{(1,2)} \cdot \underline{y}^{ik} = (N-2) (\kappa)_l; w, \quad {}^2\hat{\underline{\gamma}} = w^2 \bar{\gamma}, \\ W = \sqrt{|\gamma|}, \quad \gamma = |\gamma_{ik}|_N, \quad w = W; n$$

(M30)

$$\widehat{\mathbb{I}}=\sum_{\alpha\equiv 1}^{\omega^4}\left(\left[\begin{array}{c}\widehat{cd}\\-a\\b\end{array}\right]+sp^2Q(\alpha);()\times\left[\begin{array}{c}\widehat{cd}\\-a\\\bar{a}b\end{array}\right]\right),\quad\alpha\equiv\begin{pmatrix}cd\\ab\end{pmatrix}$$

(M31)

$$\varsigma_{klm}^i=\frac{1}{\alpha_l}l\left[\begin{array}{c}i\\kl\end{array}\right]-\frac{1}{\alpha_m}\partial_m\left[\begin{array}{c}i\\kl\end{array}\right]+\left[\begin{array}{c}i\\sl\end{array}\right];()\left[\begin{array}{cc}s&\\k&m\end{array}\right]-\left[\begin{array}{c}i\\s\\l\end{array}\right];()\left[\begin{array}{cc}s&\\k&kl\end{array}\right]$$

$$\lim_{\tau\rightarrow 0}{}^4\bar{\zeta};n={}^4\bar{R}$$

(M31a)

$${}^4\bar{\zeta}=K;\mathbb{I}$$

(M31b)

$$K_p^{(m)}=\left|\begin{array}{c}\delta_m^p,1\\(),\delta_p^m()\end{array}\right|\left|\begin{array}{c}{}^{(1,6)}\\()_{(-)p}\end{array}\right|$$